732A98 Visualization
Division of Statistics and Machine Learning
Department of Computer and Information Science

# Computer lab 2

## Instructions

- The lab report should be submitted via LISAM before the deadline. The submission should be an RMD file, HTML file and all supporting files compressed into one ZIP archive.
- Create a report for the lab solutions in RMarkdown. Make sure that it can be compiled to HTML and that all paths in RMD file are relative to the current directory where the RMD file is located. **Reports that can not be compiled are returned without revision.**
- **Include all your codes as an appendix into your report; Also show the corresponding parts of the code in the flowing text of the report.**
- A typical lab report should contain 2-4 pages of text plus some figures and the appendix with the code.
- Be concise and do not include unnecessary printouts and figures produced by the software that are not required in the assignments.

## Assignment 1

The file **olive.csv** contains information about contents of olive oils coming from different regions in Italy. Each observation contains information about

- Region (1=North, 2=South, 3= Sardinia island)
- Area (different Italian regions)
- Different acids (e.g., Palmitic, Eicosenoic, …)

1. Create a scatterplot in Ggplot2 that shows the dependence of Palmitic on Oleic in which observations are colored by Linoleic. Create also a similar scatter plot in which you divide the Linoleic variable into four classes (use cut_interval() ) and map the discretized variable to color instead. How easy/difficult is it to analyze each of these plots? What kind of perception problem is demonstrated by this experiment?
2. Create scatterplots of Palmitic vs Oleic in which you map the discretized Linoleic with four classes to:
    a. Color
    b. Size
    c. Orientation angle (use geom_spoke() )

    State in which plots it is more difficult to differentiate between the categories and connect your findings to perception metrics (i.e. how many bits can be decoded by specific aesthetics)
3. Create a scatterplot of Oleic vs Eicosenoic in which the color is defined by numeric values of Region. What is wrong with such a plot? Now create a similar kind of plot in which Region is a categorical variable. How quickly can you identify decision boundaries? Does the preattentive or attentive mechanism make it possible?
4. Create a scatterplot of Oleic vs Eicosenoic in which color is defined by discretized values of Linoleic (3 classes), shape is defined by discretized values of Palmitic (3 classes) and size is defined by discretized values of Palmitoleic (3 classes). How difficult is it to differentiate

between 27=3*3*3 different types of observations? What kind of perception problem is demonstrated by this graph?

5. Create a scatterplot of Oleic vs Eicosenoic in which color is defined by region, shape is defined by discretized values of Palmitic (3 classes) and size is defined by discretized values of Palmitoleic (3 classes). Why is it possible to clearly see a decision boundary between regions despite using many aesthetics? Explain this phenomenon from the perspective of Treisman's theory.
6. Use Plotly to create a pie chart that shows the proportions of oils coming from different Areas. Hide labels in this plot and keep only the hover-on labels. Which problem is demonstrated by this graph?
7. Create a 2D-density contour plot with Ggplot2 in which you show the dependence of Linoleic vs Eicosenoic. Compare the graph to a scatterplot with the same variables and comment on why the contour plot can be misleading.

# Assignment 2

The data set ***baseball-2016.xlsx*** contains information about the scores of baseball teams in USA in 2016, such as:

Games won, Games Lost, Runs per game, At bats, Runs, Hits, Doubles, Triples, Home runs, Runs batted in, Bases stolen, Time caught stealing, Bases on Balls, Strikeouts, Hits/At Bats, On Base Percentage, Slugging percentage, On base+Slugging, Total bases, Double plays grounded into, Times hit by pitch, Sacrifice hits, Sacrifice flies, Intentional base on balls, and Runners Left On Base.

1. Load the file to R and answer whether it is reasonable to scale these data to perform a multidimensional scaling (MDS).
2. Write R code that performs non-metric MDS with Minkowski distance=2 of the data (numerical columns) into two dimensions. Visualize the resulting observations in Plotly as a scatter plot in which observations are coloured by League. Does a difference between the leagues exist according to the plot? Which of the MDS components seem to provide the best differentiation between the Leagues? Which baseball teams seem to be outliers?
3. Use Plotly to create a Shepard plot for the MDS performed and comment about how successful the MDS was. Which observation pairs were hard for the MDS to map successfully?
4. Produce a series of scatterplots in which you plot the MDS variable that was the best in the differentiation between the leagues in step 2 against all other numerical variables of the data. Pick up two scatterplots that seem to show the strongest (positive or negative) connection between the variables and include them into your report. Find some information about these variables in Google – do they appear to be important in scoring the baseball teams? Provide some interpretation for the chosen MDS variable.

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# Submission procedure

When submitting the report, remember to **specify in LISAM your group name** and all group members that wrote the report!

Make sure that you or your group mate submits the group report using *Lab X* item in the *Submissions* folder before the deadline. Make sure that **the report contains a Statement of Contribution** describing how each group member has contributed into the group report.

